Assignment 9

1.In recent years the interest rate on home mortgages has declined to less than 6.0 percent. However, according to a study by Federal Reserve Board the rate charge on credit card debit is more than 14 percent. Listed below is the interest rate charged on a sample of 10 credit cards.

14.6	16.7	17.4	17.0	17.8	15.4	13.1	15.8
14.3	14.5						

Is it reasonable to conclude the mean rate charged is greater than 14 percent? Use the 0.01 significance level.

$$\bar{X} = \frac{14.6+16.7+\cdots+14.5}{10} = 15.66$$

$$S = \sqrt{\frac{\sum(x_i-\bar{x})^2}{n-1}} = \sqrt{\frac{(14.6-15.66)^2+(16.7-15.66)^2+\cdots+(145-15.66)^2}{9}} = 1.5$$

1. $H_0: u \le 14$ versus $H_a: u > 14$

2. $\alpha = 0.01$, $n = 10$

3. This is a t test

4. For $\alpha = 0.01$, the critical t value is 2.821

5. $t = \frac{\bar{X}-u}{S/\sqrt{n}} = \frac{15.66-14}{1.54/\sqrt{10}} = 3.409$

6. Since $t = 3.409 > 2.821$, reject H_0 at 0.01.

It is reasonable to conclude the mean rate change is greater than 14 percent.

2. The Computer Anxiety Rating Scale (CARS) measures an individual's level of computer anxiety on a scale from 20 (no anxiety) to 100 (highest level of anxiety). Researchers at Miami University administered CARS to 172 business students. One of the objectives of the study was to determine if there is a difference between the level of computer anxiety experienced by female and male business students. We assume that variances of two populations are equal.

	Males	Females
$\bar{x}$	40.26	36.85
s	13.35	9.42
n	100	72

At the 0.05 level of significance, is there evidence of a difference in the mean computer anxiety experienced by female and male business students?

Given: $n_1 = 100$ $\bar{X}_1 = 40.26$ $S_1 = 13.35$

$n_2 = 72$ $\bar{X}_2 = 36.85$ $S_2 = 9.42$

1. $H_0: u_1 = u_2$ versus $H_a: u_1 \neq u_2$

2. $\alpha = 0.05$

3. Since σ_1 and σ_2 unknown, assumed equal.

this is a t test with $df = n_1 + n_2 - 2 = 170$

4. For $\alpha = 0.05$, the critical t values are ± 1.96

5. $$S_p^2 = \frac{(n_1-1)S_1^2 + (n_2-1)S_2^2}{(n_1-1)+(n_2-1)} = \frac{(100-1)(13.35)^2 + (72-1)(9.42)^2}{(100-1)+(72-1)} = 140.85$$

$$t = \frac{(\bar{X}_1 - \bar{X}_2) - (u_1 - u_2)}{\sqrt{S_p^2\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} = \frac{(40.26 - 36.85) - 0}{\sqrt{140.85\left(\frac{1}{100} + \frac{1}{72}\right)}} = 1.859$$

6. Since $t = 1.859 < 1.96$, do not reject H_0

There is no sufficient evidence of a difference in the mean computer anxiety experienced by female a male business Students

3. A number of minor automobile accidents occur at various high-risk intersections in Teton County despite traffic lights. The Traffic Department claims that a modification in the type of light will reduce these accidents. The county commissioners have agreed to a proposed experiment. Eight intersections were chosen at random, and the lights at those intersections were modified. The numbers of minor accidents during a six-month period before and after the modifications were:

	Number of Accidents							
	A	B	C	D	E	F	G	H
Before	5	7	6	4	8	9	8	10
After	3	7	7	0	4	6	8	2

At the 0.05 level of significance, is there evidence that the modification in the type of light will reduce the accidents?

$d_1 = 5-3 = 2.\ d_2 = 0\ \ d_3 = -1,\ d_4 = 4.\ d_5 = 4.\ d_6 = 3.\ d_7 = 0.\ d_8 =$

$$\bar{d} = \frac{d_1 + d_2 + \cdots + d_8}{n} = \frac{2 + 0 + \cdots + 8}{8} = 2.5$$

$$S_d = \sqrt{\frac{\sum_{i=1}^{n}(d_i - \bar{d})^2}{n-1}} = \sqrt{\frac{(2-2.5)^2 + \cdots + (8-2.5)^2}{7}} = 2.928$$

1. $H_0: \mu_d = 0$ versus ~~Ha~~ $H_a: \mu_d \neq 0$

2. $\alpha = 0.05$

3. This is a paired t test, with $df = n-1 = 8-1 = 7$

4. For $\alpha = 0.05$, the critical t value is ± 2.365.

5. $$t = \frac{\bar{d} - \mu_d}{S_d/\sqrt{n}} = \frac{2.5 - 0}{2.928/\sqrt{8}} = 2.415$$

6. Since $t = 2.415 > 2.365$. reject H_0 at 0.05

There is evidence that the modification in the type of light will reduce the accidents.